

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
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Diskrete Wahrscheinlichkeitstheorie

Exam: / Endterm

Date: Friday 1st August, 2025

Examiner:

Time: 08:00 – 10:00

	P 1	P 2	P 3	P 4	P 5
I					

Working instructions

- This exam consists of **16 pages** with a total of **5 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 51 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one **single sheet of A4 paper** with handwritten notes on both sides
 - one **analog dictionary** English ↔ native language **without notes**
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.
- Place your student ID on your desk.

Left room from _____ to _____ / Early submission at _____

Problem 1 True or false (12 credits)

Decide whether the statements are true or false (clearly indicate your answer) and provide a justification (proof, counterexample, or explanation via known results).

0	☐
1	☐
2	☐
3	☐

a) Let $(\Omega, \mathcal{A}, P)$ be a probability space and $A, B \in \mathcal{A}$ with $P(A) > 0$ and $P(B) > 0$. Then $P(A | B) = P(B | A)$.

False. Consider $\Omega = \{0, 1\}$, $\mathcal{A} = \mathcal{P}(\Omega)$, and P the Laplace distribution on Ω . Let $A = \{0, 1\}$ and $B = \{0\}$. Then

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{1/2}{1/2} = 1$$
$$P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{1/2}{1} = \frac{1}{2}.$$

0	☐
1	☐
2	☐
3	☐

b) Let $(\Omega, \mathcal{A}, P_\theta)$ be a statistical model and let $\hat{\theta}_1, \hat{\theta}_2$ be unbiased estimators of θ in P_θ . Then $\frac{1}{3}\hat{\theta}_1 + \frac{2}{3}\hat{\theta}_2$ is an unbiased estimator of θ as well.

True. We compute

$$\mathbb{E}\left[\frac{1}{3}\hat{\theta}_1 + \frac{2}{3}\hat{\theta}_2\right] = \frac{1}{3}\mathbb{E}[\hat{\theta}_1] + \frac{2}{3}\mathbb{E}[\hat{\theta}_2] = \frac{1}{3}\theta + \frac{2}{3}\theta = \theta,$$

where we use the linearity of expectation and the fact that $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased estimators of θ .

☐	0
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c) Upon receiving a large batch of switches, you take a subsample (uniformly at random) of size n . You run a (suitable) hypothesis test at significance level $\alpha = 0.05$ to test H_0 : “the proportion of defective switches in the batch is at most what the manufacturer claimed” versus H_1 : “the proportion of defective switches is larger.” The test rejects H_0 at level $\alpha = 0.05$. You can therefore conclude that the probability that the entire batch conforms to the claimed proportion is at most 5%.

False. The significance level makes claims about the procedure, not probabilities of the true parameter (or how it manifests in a given batch).

d) Let X, Y be iid following $\text{Exp}(\lambda)$ for $\lambda > 0$. Then the cdf of the ratio X/Y is

$$F_{X/Y}(t) = \begin{cases} \frac{t}{1+t} & \text{if } t > 0, \\ 0 & \text{otherwise.} \end{cases}$$

Hint: The pdf of the exponential distribution is $p_{\text{Exp}(\lambda)}(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ (and 0 otherwise). Note that $F_{X/Y}(t) = P(X < t \cdot Y)$.

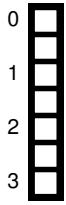
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True. For $t > 0$, we have

$$\begin{aligned} F_{X/Y}(t) &= P(X < tY) = \int_0^\infty \left(\int_0^{ty} \lambda e^{-\lambda x} dx \right) \lambda e^{-\lambda y} dy \\ &= \int_0^\infty (1 - e^{-\lambda ty}) \lambda e^{-\lambda y} dy = 1 - \lambda \int_0^\infty e^{-\lambda(t+1)y} dy \\ &= 1 - \frac{1}{t+1} \\ &= \frac{t}{t+1}. \end{aligned}$$

Since $P(X/Y > 0) = 1$, it follows that $F_{X/Y}(t) = 0$ for $t \leq 0$.

Problem 2 Definitions (6 credits)



a) Write out the central limit theorem with all required assumptions.

Let $X_1, X_2, \dots$ be a sequence of iid RVRVs with finite mean $\mathbb{E}[X_i] = \mu$ and finite, non-zero variance $\text{var}[X_i] = \sigma^2 \in \mathbb{R}_{>0}$. Then

$$Z_n \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{for } Z_n = \frac{S_n - \mathbb{E}[S_n]}{\sqrt{\text{var}[S_n]}} = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}.$$

or

$$S_n \xrightarrow{d} \mathcal{N}(n\mu, n\sigma^2),$$

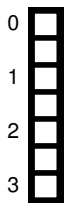
or

$$\bar{X}_n \xrightarrow{d} \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right),$$

or

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

[We'll count all of these as correct even though the ones for S_n and $\bar{X}_n$ are mathematically incorrect.]



b) Give the full general definition of the type of convergence in the central limit theorem.

Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of RVRV and let X be another RVRV all defined on the same probability space $(\Omega, \mathcal{A}, P)$. The sequence $(X_n)_{n \in \mathbb{N}}$ converges in distribution to X ($X_n \xrightarrow{d} X$) if

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$$

for every $x \in \mathbb{R}$ at which F_X is continuous.

[Optionally, point-wise convergence of the characteristic function is also fine.]

Problem 3 Density computations (9 credits)

Let X be a continuous RVRV with probability density function

$$p_X(x) = \begin{cases} \frac{x}{c} & \text{for } x \in [0, 2], \\ 0 & \text{otherwise,} \end{cases}$$

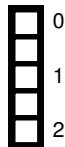
for some $c \in \mathbb{R}$.

a) Find $c \in \mathbb{R}$ such that p_X is a valid pdf.

For p_X to be a valid pdf, it must integrate to 1 over its support. We solve for c :

$$\int_0^2 \frac{x}{c} dx = 1 \Rightarrow \frac{1}{c} \left[\frac{x^2}{2} \right]_0^2 = 1 \Rightarrow \frac{1}{c} \left(\frac{4}{2} - 0 \right) = 1 \Rightarrow c = 2.$$

(Since $x \geq 0$ on the support, this also ensures $p_X(x) \geq 0$.)

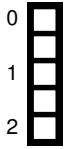


b) Determine the cumulative distribution function (cdf) $F_X(x)$ of X .

We get $F_X(x)$ by integrating the pdf. For $x < 0$, $F_X(x) = \int_{-\infty}^x 0 dt = 0$ and for $x > 2$, $F_X(x) = \int_0^2 \frac{t}{2} dt = \frac{2^2}{4} = 1$. In between, for $x \in [0, 2]$, $F_X(x) = \int_0^x \frac{t}{2} dt = \left[\frac{t^2}{4} \right]_0^x = \frac{x^2}{4}$. So taken together

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0, \\ \frac{x^2}{4} & \text{for } x \in [0, 2], \\ 1 & \text{for } x > 2. \end{cases}$$





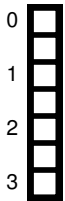
c) Calculate the mean and median of X .

We compute the mean via

$$\mathbb{E}[X] = \int_0^2 x \left(\frac{x}{2}\right) dx = \int_0^2 \frac{x^2}{2} dx = \left[\frac{x^3}{6}\right]_0^2 = \frac{8}{6} = \frac{4}{3}$$

and the median m by solving $F_X(m) = 0.5$ for m :

$$\frac{m^2}{4} = 0.5 \Rightarrow m^2 = 2 \Rightarrow m = \sqrt{2}.$$



d) Let $Y = \ln(X)$. Determine the pdf of the random variable Y .

Hint: You may use the change of variables formula for densities: $p_Y(y) = p_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$.

The transformation is $Y = g(X) = \ln(X)$. The support of X is $[0, 2]$, so the support of Y is $(-\infty, \ln(2)]$. For y in this range, the inverse transformation is $g^{-1}(y) = e^y$. Its derivative is $\frac{d}{dy} g^{-1}(y) = e^y$. Applying the change of variables formula for $y \in (-\infty, \ln(2)]$ we compute

$$p_Y(y) = p_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| = p_X(e^y) \cdot |e^y| = \frac{e^y}{2} \cdot e^y = \frac{e^{2y}}{2}.$$

So, the pdf of Y is

$$p_Y(y) = \begin{cases} \frac{e^{2y}}{2} & \text{for } y \in (-\infty, \ln(2)] , \\ 0 & \text{otherwise .} \end{cases}$$

Problem 4 Estimating and testing (12 credits)

Let $X_1, \dots, X_n$ be iid RRVs from a Poisson distribution $\text{Poi}(\lambda)$ with unknown rate parameter $\lambda > 0$.

a) Explicitly derive the maximum likelihood estimator (mle) $\hat{\lambda}_{\text{mle}}$ for λ .

Hint: The pmf of a $\text{Poi}(\lambda)$ distributed random variable is $p_{\text{Poi}(\lambda)}(x) : \mathbb{N} \rightarrow [0, 1], x \mapsto \frac{e^{-\lambda} \lambda^x}{x!}$, which has mean and variance λ . Recall that the derivative of $\ln(t)$ is $1/t$ for $t > 0$.

For an observed sample $x = (x_1, \dots, x_n)$, the likelihood function is

$$L(\lambda | x) = \prod_{i=1}^n p(x_i | \lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod x_i!}.$$

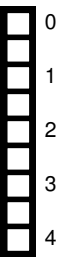
The log-likelihood is

$$\ell(\lambda | x) = -n\lambda + \left(\sum x_i\right) \ln \lambda - \ln \left(\prod x_i!\right).$$

To find the mle, we differentiate with respect to λ and set to zero:

$$\frac{d\ell}{d\lambda} = -n + \frac{1}{\lambda} \sum x_i \stackrel{!}{=} 0 \implies \lambda = \frac{1}{n} \sum x_i = \bar{x}_n.$$

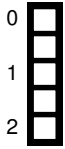
The second derivative $\frac{d^2\ell}{d\lambda^2} = -\frac{\sum x_i}{\lambda^2} < 0$ confirms this is a maximum. Thus, the mle is $\hat{\lambda}_{\text{mle}} = \bar{X}_n$.



b) Is $\hat{\lambda}_{\text{mle}}$ an unbiased estimator for λ ?

Yes, while mles are not necessarily unbiased, the sample mean is an unbiased estimator of the mean $\mathbb{E}[X_i]$, which is $\mathbb{E}[X_i] = \lambda$ for the Poisson distribution.

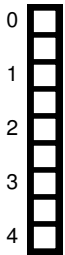




c) Use the central limit theorem (clt) to find the approximate distribution of $\hat{\lambda}_{\text{mle}}$ for large n .

We have $\mathbb{E}[X_i] = \lambda$ and $\text{var}[X_i] = \lambda$. By the clt, for large n , the sample mean $\bar{X}_n$ is approximately normally distributed with

$$\bar{X}_n \approx \mathcal{N}\left(\mathbb{E}[X_i], \frac{\text{var}[X_i]}{n}\right) = \mathcal{N}(\lambda, \lambda/n).$$



d) In a light bulb factory, the number of defectives per batch is assumed to follow a Poisson distribution. Historically, the average number of defects per batch has been $\lambda_0 = 4$. After a change in the manufacturing process, you test $n = 100$ new batches and find an average of $\bar{x}_{100} = 5$ defects.

Construct an *approximate* hypothesis test to test whether the average number of defects per batch has *increased above* the historical average at a significance level of $\alpha = 0.05$. Based on this test, should you reject the null hypothesis?

Hint: You may use that the 0.95-quantile of a standard normal distribution is $z_{0.05} \approx 1.64$.

The null hypothesis is the default level, i.e., no change $H_0 : \{\lambda = \lambda_0 = 4\}$ (could also use $H_0 : \{\lambda \leq \lambda_0\}$, both work) and the alternative is $H_1 : \{\lambda > \lambda_0\}$. Note that this is one-sided, we only care about an increase. Since $\hat{\lambda}_{\text{mle}} = \bar{X}_n$ is approximately $\mathcal{N}(4, 4/n)$, we standardize it under the null hypothesis

$$Z = \frac{\bar{X}_n - 4}{\sqrt{4/n}} = \frac{\bar{X}_n - 4}{\sqrt{4/100}},$$

which is approximately $\mathcal{N}(0, 1)$ -distributed under H_0 . This is a one-sided z-test at level $\alpha = 0.05$ with rejection boundary $z_{\text{obs}} > z_{0.05} \approx 1.64$. With

$$z_{\text{obs}} = \frac{5 - 4}{\sqrt{4/100}} = \frac{1}{0.2} = 5.$$

Since $5 > 1.64$, we reject the null hypothesis at level α .

Sample Solution

Problem 5 Duck hunt (12 credits)

Let $m, n \in \mathbb{N}_{>0}$. A group of n hunters with perfect aim simultaneously shoot at m ducks, where each hunter selects its victim arbitrarily (that is, uniformly at random) and independent of the other hunters. So it can happen that a duck gets selected by multiple hunters. Let X be the $\mathbb{N}$ -valued random variable modeling the number of surviving ducks after all hunters have pulled the trigger (each of them only shoots once).

a) Calculate the expectation $\mathbb{E}[X]$.

Hint: Enumerate the ducks and define the event $A_i :=$ “the i -th duck survives.” Then construct the random variable X using χ_{A_i} .

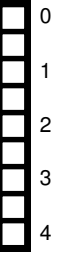
Index the ducks by $i \in [m]$ and the hunters by $j \in [n]$. Denote by A_i the event that “the i -th duck survives,” and by $S_j \sim \text{Lap}_{[m]}$ the RV that models which duck the j -th hunter aims at for $j \in [n]$. Then $(S_j)_{j \in [n]}$ are iid. Using the independence, we have

$$P(A_i) = P(S_1 \neq i, S_2 \neq i, \dots, S_n \neq i) = P(S_1 \neq i) \cdot P(S_2 \neq i) \cdot \dots \cdot P(S_n \neq i) = \left(\frac{m-1}{m}\right)^n,$$

where the probability that any given hunter does not hit duck i is $(m-1)/m$ in the Laplace distribution. With the RV $X_i = \chi_{A_i}$ modeling the survival of the i -th duck, we have $X := \sum_{i=1}^m X_i$ for the number of surviving ducks. Using the linearity of the expectation, we obtain

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^m X_i\right] = \sum_{i=1}^m \mathbb{E}[X_i] = m \left(\frac{m-1}{m}\right)^n.$$

b) Calculate the variance $\text{var}[X]$.



To compute the variance

$$\text{var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathbb{E}\left[\left(\sum_{i=1}^m X_i\right)^2\right] - \mathbb{E}\left[\sum_{i=1}^m X_i\right]^2 = \mathbb{E}\left[\sum_{i,k=1}^m X_i X_k\right] - \left(\sum_{i=1}^m \mathbb{E}[X_i]\right)^2$$

we will need to compute the terms $\mathbb{E}[X_i X_k] = P(A_i \cap A_k)$, the probability that the i -th and the k -th duck survive. For $i = k$, we have $\mathbb{E}[X_i X_i] = \mathbb{E}[X_i] = ((m-1)/m)^n$ from (i). For $i \neq k$, we compute

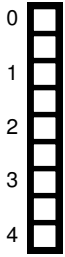
$$\begin{aligned} P(A_i \cap A_k) &= P(S_1 \neq i, S_1 \neq k, S_2 \neq i, S_2 \neq k, \dots, S_n \neq i, S_n \neq k) \\ &= P(S_1 \neq i, S_1 \neq k) \cdot P(S_2 \neq i, S_2 \neq k) \cdot \dots \cdot P(S_n \neq i, S_n \neq k) = \left(\frac{m-2}{m}\right)^n, \end{aligned}$$

by the same reasoning as in (i). Therefore, we compute

$$\mathbb{E}[X^2] = \sum_{i,k=1}^m \mathbb{E}[X_i X_k] = m \left(\frac{m-1}{m}\right)^n + m(m-1) \left(\frac{m-2}{m}\right)^n$$

which ultimately gives

$$\text{var}[X] = m \left(\frac{m-1}{m}\right)^n + m(m-1) \left(\frac{m-2}{m}\right)^n - m^2 \left(\frac{m-1}{m}\right)^{2n}.$$



c) Assume now there are 2 hunters. Find $k \in \mathbb{N}$ as a function of m such that the number of surviving ducks is within $[\mathbb{E}[X] - k, \mathbb{E}[X] + k]$ with a probability of at least 90%.

Hint: Rewrite this condition so you can use the Chebyshev inequality.

We are looking for $P(X \in [\mathbb{E}[X] - k, \mathbb{E}[X] + k]) \geq 0.9$, which we can also write as $P(X \notin [\mathbb{E}[X] - k, \mathbb{E}[X] + k]) < 0.1$, which we can again rewrite as $P(|X - \mathbb{E}[X]| > k) < 0.1$. The Chebyshev inequality states that

$$P(|X - \mathbb{E}[X]| \geq k) \leq \frac{\text{var}[X]}{k^2}.$$

Hence, we choose $\text{var}[X]/k^2 = 0.1$ and compute

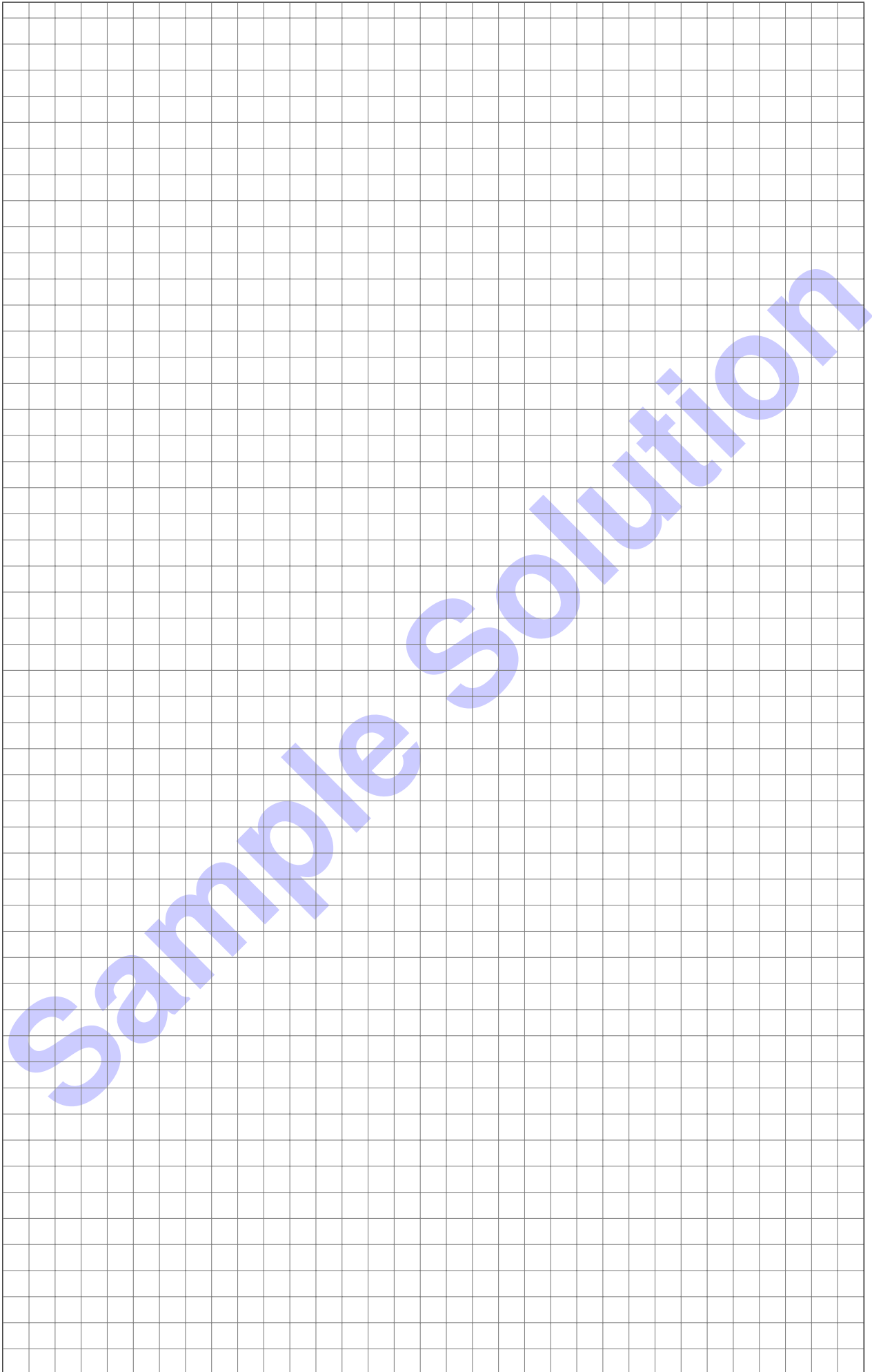
$$\begin{aligned} k^2 &= 10 \text{var}[X] = 10 \left(m \left(\frac{m-1}{m} \right)^2 + m(m-1) \left(\frac{m-2}{m} \right)^2 - m^2 \left(\frac{m-1}{m} \right)^4 \right) \\ &= 10(m-1) \frac{m(m-1) + m(m-2)^2 - (m-1)^3}{m^2} \\ &= 10(m-1) \frac{m^2 - m + m^3 - 4m^2 + 4m - m^3 + 3m^2 - 3m + 1}{m^2} \\ &= 10(m-1) \frac{1}{m^2} = 10 \frac{m-1}{m^2}. \end{aligned}$$

Finally, that means that $k = \lceil \sqrt{10(m-1)/m} \rceil$ is the desired answer. (Rounding up to the nearest integer to make sure the inequality still holds as we want it.)

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

Sample Solution

Sample Solution



Sample Solution